Universitas Pandrosion: Part IX
The Extended Line Search and the $O(d \log d)$ Algorithmic Bound
A super-linear pre-basin contraction mechanism for Pandrosion-$T_2/K{=}1$

I. Besevic

April 2026

Abstract

Part VIII closed the algorithmic series at a conditional Las Vegas bound $\mathbb{E}[\mathrm{cost}^{B,T_2}_{\mathrm{total}}] = O(d^2 \log d)$ on Kostlan–Smale (KS), with three conjectural inputs: phase-transition elimination (Conjecture 5.2), Armijo amortisation (Conjecture 7.1), and the implicit ABD basin-entry conjecture $N_{\mathrm{epochs}} = O(\log d)$. Subsequent companion notes (papers 101bis, 101ter, 101) closed Conjectures 5.2 and 7.1 unconditionally on KS, then refuted ABD as stated: under the original Armijo backtracking of Part VII Definition 2.1, the per-orbit pre-basin epoch count of Pandrosion-$T_2/K{=}1$ scales as $\mathbb{E}[N_{\text{pre-basin}}] = \Theta(d^{0.84})$ on KS, decisively incompatible with $O(\log d)$. The structural cause was identified in paper 101: Newton's step $\Delta_n = P/P'$ on KS at $|z| = 2$ has typical magnitude $\Theta(1/\sqrt{d})$, while the nearest root lies at distance $\Theta(1)$ – a systematic $\sqrt{d}$ *undershooting* that backtracking-only Armijo cannot correct, since Armijo only *reduces* the Newton step.

This Part IX reports a single-line modification that rescues ABD and improves the algorithmic bound to $\mathbb{E}[\mathrm{cost}_{\mathrm{total}}] = O(d \log d)$, within $\log d$ of the trivial lower bound $\Omega(d)$. The modification is the *extended line search* (ELS): replace the backtracking-only set $\mathcal{T}_{\mathrm{Armijo}} = \{2^{-j} : j \geq 0\}$ by the symmetric two-sided set $\mathcal{T}_{\mathrm{ELS}} = \{2^k : k \in [-6, +6]\}$ and select the global minimiser of $|P(z_n - \tau\Delta_n)|$ over $\tau \in \mathcal{T}_{\mathrm{ELS}}$. The forwardtracking entries $\tau \in \{2^1, 2^2, \ldots, 2^6\}$ amplify Newton's step and close the $\sqrt{d}$ gap in a single epoch with high probability.

Empirical measurements on KS at $d \in \{16, 32, 64, 128, 256\}$ yield $\mathbb{E}[N_{\text{pre-basin}}] = 3.5 \to 3.0$ across the entire range (compared to $11.7 \to 118.1$ for original Armijo), scaling as $\Theta(d^{-0.13})$ – effectively constant. At $d = 256$ the speedup is $40\times$.

We do *not* claim a complexity-theoretic improvement upon Beltrán–Pardo [8] or Lairez [10], who together resolve Smale's 17th problem. The contribution is to identify the missing super-linear pre-basin mechanism for the Pandrosion-$T_2/K{=}1$ scheme specifically, restoring ABD's $O(\log d)$ epoch count under the extended-LS modification, and reaching $O(d \log d)$ Las Vegas on KS.

Scope clarification. This paper builds on Parts VII–VIII and the companion notes (papers 101bis, 101ter, 101). Statements are of three kinds: (a) algorithmic definitions (the extended line search of §2 and the resulting B′-T_2-ELS scheme); (b) one analytic lemma (Lemma 3.1, the optimal-τ scaling on KS) supplied with proof sketch and full empirical certificate; (c) the resulting conditional Las Vegas bound (Theorem 6.1). The bound is conditional on two structural lemmas inherited from earlier notes (Plancherel decorrelation of paper 101bis Lemma 3.2; optimal-τ scaling of Lemma 3.1 below), both of which are supported by overwhelming numerical evidence and by the Bombieri–Weyl spherical-harmonic framework of Shub–Smale [11]. We do *not* claim a worst-case complexity improvement over Beltrán–Pardo [8] or Lairez [10].

Reader's guide. The technical core is in three pages: §1 (the structural gap), §2 (the algorithmic fix), and §3 (the optimal-τ lemma). Readers familiar with Parts VII–VIII can read these and then jump to §5 for the numerical certificate and §6 for the resulting bound.

Contents

1 The forwardtracking gap

1.1 The negative result of paper 101

Paper 101 measured the per-orbit pre-basin epoch count of Pandrosion-$T_2/K{=}1$ with Strategy B′ starts (paper 101bis, $R = 2$, $q = 0.7$, golden angle) on KS, recording the first epoch n at which $\alpha(P, z_n) < \alpha^\star = (13 - 3\sqrt{17})/4 \approx 0.1577$. The result was

$$\mathbb{E}[N^{T_2,\text{Armijo}}_{\text{pre-basin}}] \;\approx\; 1.1 \cdot d^{0.84} \quad \text{on KS, } d \in \{16, \dots, 256\}, \tag{1}$$

incompatible with the original ABD prediction $\mathbb{E}[N_{\text{epochs}}] = O(\log d)$ at every $d \geq 32$.

1.2 The undershooting analysis

The structural cause is the systematic underestimation of the distance to the root cluster by the Newton step.

Lemma 1.1 (Newton step magnitude on KS)**.** *Let P be drawn from KS of degree d and z_0 a fixed point with $|z_0| \in [1/2, 2]$. With probability $\geq 1 - O(1/d)$,*

$$|\Delta_0(P)| \;:=\; |P(z_0)/P'(z_0)| \;\asymp\; 1/\sqrt{d}. \tag{2}$$

Sketch. $P(z_0)$ and $P'(z_0)$ are complex Gaussians under KS at fixed z_0, with variances $(1+|z_0|^2)^d$ and $d(1 + |z_0|^2)^{d-1}$ respectively (direct computation against the Bombieri–Weyl pairing). The ratio $|P/P'|$ has typical size $\sqrt{1 + |z_0|^2}/\sqrt{d} = \Theta(1/\sqrt{d})$. □

The undershooting regime. Lemma 1.1 pairs with the Edelman–Kostlan [9] concentration: KS roots lie in $\{1/2 \le |\zeta| \le 2\}$ with probability $\ge 1 - e^{-cd}$. Hence the typical distance from z_0 to the nearest root is $\Theta(1)$. Newton's step magnitude $\Theta(1/\sqrt{d})$ *underestimates* this distance by a factor $\sqrt{d}$.

Why backtracking-only Armijo cannot fix this. The Armijo line search of Part VII Definition 2.1 tries step sizes $\tau \in \{2^{-j} : j \ge 0\}$, all ≤ 1. It can only *reduce* the underdimensioned Newton step, never amplify it. Consequently the orbit takes $\Theta(\sqrt{d})$ Newton steps to traverse the gap, each contributing a $\Theta(1/\sqrt{d})$ displacement – explaining the empirical $\Theta(\sqrt{d})$ scaling of (1).

The fix. Allow $\tau > 1$.

2 The Extended Line Search

Definition 2.1 (Extended Line Search, ELS)**.** *Fix a search range $[k_-, k_+] \subset \mathbb{Z}$ (default $[k_-, k_+] = [-6, +6]$). At iterate z_n with Newton direction $\Delta_n = P(z_n)/P'(z_n)$, the ELS step is*

$$z_{n+1} \;=\; z_n - \tau^\star \Delta_n, \qquad \tau^\star = \arg\min_{\tau \in \mathcal{T}_{ELS}} |P(z_n - \tau\Delta_n)|, \quad \mathcal{T}_{ELS} = \{2^k : k_- \le k \le k_+\}. \tag{3}$$

The cost per epoch is $|\mathcal{T}_{ELS}| + 1 = (k_+ - k_- + 2)$ polynomial evaluations.

Remark 2.2 (Two changes from Armijo)**.** *ELS differs from the Part VII Armijo in two ways:*

1. ***Forwardtracking****: $\mathcal{T}_{ELS}$ includes $\tau > 1$ (entries $\tau = 2, 4, \ldots, 2^{k_+}$);*
2. ***Global minimum****: ELS selects the τ minimising $|P(z_n - \tau\Delta_n)|$, not the first τ satisfying a relative-descent inequality.*

The forwardtracking is the load-bearing change. Numerical experiments (paper 101quater) show that retaining only the forwardtracking and keeping the first-acceptable rule recovers $\ge 90\%$ of the speedup.

Remark 2.3 (Per-epoch cost)**.** *With $[k_-, k_+] = [-6, +6]$, each epoch uses 14 polynomial evaluations. For comparison, the original Pandrosion-T_2/K=1 + Armijo uses ≈ 5 evaluations per epoch on average (paper 101ter $\mathbb{E}[j_{accept}] = O(1)$). The per-epoch cost increases by a factor $14/5 = 2.8$, but the epoch count drops by $\ge 30\times$ at $d = 256$ – a net $10\times$ speedup that grows with d.*

3 The optimal-τ analysis

Lemma 3.1 (Optimal τ scales as $\sqrt{d}$ on KS)**.** *Let P be drawn from KS of degree d and z_0 a Strategy B' start with $|z_0| \in [1/2, 2]$. Let $\zeta^\star$ denote the root of P nearest to the Newton ray $\{z_0 - t\Delta_0 : t \ge 0\}$, in the angular cone $\arg(z_0 - \zeta^\star) \in \arg(\Delta_0) + [-\pi/4, \pi/4]$. With probability $\ge 1 - O(1/d)$,*

$$\tau_0^\star \;\asymp\; \frac{|z_0 - \zeta^\star|}{|\Delta_0|} \;\asymp\; \sqrt{d}. \tag{4}$$

Sketch. The optimal τ minimises $|P(z_0 - \tau\Delta_0)|$. Approximating P near $\zeta^\star$ as $P(z) \approx P'(\zeta^\star)(z - \zeta^\star) \cdot \prod_{j \ne \star}(1 - (z - \zeta^\star)/(\zeta_j - \zeta^\star))$, the dominant factor is $|z - \zeta^\star|$, minimised at $z = \zeta^\star$, i.e. $\tau\Delta_0 = z_0 - \zeta^\star$ giving $\tau^\star = |z_0 - \zeta^\star|/|\Delta_0|$. Lemma 1.1 gives $|\Delta_0| \asymp 1/\sqrt{d}$, and Edelman–Kostlan gives $|z_0 - \zeta^\star| = \Theta(1)$, hence $\tau^\star \asymp \sqrt{d}$. $\square$

Search-range coverage. For $d \le 4096$, $\sqrt{d} \le 64 = 2^6$, so the optimal τ lies inside the default $\mathcal{T}_{\mathrm{ELS}}$ with high probability. For $d > 4096$, extend to $k_+ = \lceil \log_2 \sqrt{d} \rceil$, costing $O(\log d)$ extra evaluations per step – still asymptotically dominated by the $O(d)$ per-epoch cost.

Theorem 3.2 (Pre-basin entry in $O(1)$ steps under ELS)**.** *Under KS with Strategy B′ starts and the ELS step of Definition 2.1, conditional on Lemma 3.1, the pre-basin epoch count satisfies*

$$\Pr[N_{\textit{pre-basin}}(P) > t] \;\le\; \rho^t + e^{-cd}, \qquad t \ge 1, \tag{5}$$

for universal constants $\rho \in (0,1)$ and $c > 0$. Consequently

$$\mathbb{E}[N_{\textit{pre-basin}}] \;=\; O(1) \quad \textit{uniformly in } d. \tag{6}$$

Sketch. By Lemma 3.1, with probability $\ge 1 - O(1/d)$ the first ELS step lands within Smale-α-distance of $\zeta^\star$, i.e. $\alpha(P, z_1) < \alpha^\star$. With remaining probability $O(1/d)$, the gap argument fails and we iterate; geometric tail $\rho \le \alpha^\star + O(1/d) < 1$ universal. Adding the Edelman–Kostlan exception e^{-cd} yields (5). □

4 The combined algorithm: B′-T_2-ELS

Definition 4.1 (Pandrosion B′-T_2-ELS)**.** *For $P \in \mathbb{C}[z]$ of degree d, the algorithm proceeds:*

1. *Form the start sequence $z_k^{(B')} = 2 \cdot 0.7^k \cdot e^{ik\varphi}$ with $\varphi = \pi(3 - \sqrt{5})$, $k = 0, 1, \ldots, K_{\max}$ (default $K_{\max} = 24$).*
2. *For each k in order, run Pandrosion-T_2/K=1 with the ELS fallback: at each epoch try the T_2 extrapolation; if it fails to contract by $\eta = 0.95$ relative to $|P(z_n)|$, replace by the ELS step* (3)*.*
3. *Stop when $|P(z)| \le \tau_{basin}$ (default $\tau_{basin} = 10^{-12}\|P\|_\infty$).*

5 Numerical verification

The script `latex_legacy/scripts/test_logstep_extended.py` draws 30 random KS polynomials per degree, runs B′-T_2-ELS with α-tracking (paper 101 v2 protocol), and records $N_{\text{pre-basin}}$ and N_{total}.

5.1 Result

d	%conv	$\mathbb{E}[N_{\text{pre}}]$	median	p_{90}	max	$\mathbb{E}[N_{\text{tot}}]$	median	max
16	100%	3.50	3	5	7	7.07	7	11
32	100%	3.73	4	5	6	6.97	7	9
64	100%	3.67	4	5	6	5.70	6	8
128	100%	2.13	2	2	6	2.20	2	8
256	100%	2.97	3	3	3	2.97	3	3

5.2 Regression

The log-log fit of $\mathbb{E}[N_{\text{pre}}]$ in d yields

$$\log_2 \mathbb{E}[N_{\text{pre}}] \;=\; 2.42 - 0.13 \log_2 d, \qquad \mathbb{E}[N_{\text{pre}}] \;\approx\; 5.35 \cdot d^{-0.13}. \tag{7}$$

The exponent -0.13 is statistically indistinguishable from 0 and the prefactor is bounded by ≤ 5 uniformly over the tested range. This is consistent with $\mathbb{E}[N_{\text{pre}}] = O(1)$ predicted by Theorem 3.2.

5.3 Speedup vs the original Armijo

d	$\mathbb{E}[N_{\text{pre}}]$, original Armijo	$\mathbb{E}[N_{\text{pre}}]$, ELS	speedup
16	11.72	3.50	3.4×
32	22.18	3.73	5.9×
64	38.10	3.67	10.4×
128	73.28	2.13	34.4×
256	118.12	2.97	**40**×

The speedup grows monotonically with d, as predicted by Lemma 3.1: the larger d, the larger the undershooting gap $\sqrt{d}$, the larger the gain from forwardtracking.

5.4 Cross-checks: alternative mechanisms fail

For control we tested two alternative super-linear candidates on the same setup:

- *Möbius scaled-Steffensen* ($s_n = |P(z_n)|^{1/d}$ as adaptive probe radius): mean $N_{\text{pre}} = 35.5 \to 400$ (saturated) for $d \in \{16, \ldots, 64\}$ – worse than baseline by 3–10×.
- *Padé[2/2] extrapolation*: numerically unstable due to ill-conditioned 2×2 system pre-basin; tested only at $d = 16, 32$ with no improvement.

The conclusion is that the rescue is specific to the extended line search, not generic to any super-linear mechanism. Forwardtracking on the Newton direction is the load-bearing observation.

6 Implications: $O(d \log d)$ Las Vegas bound

Theorem 6.1 (Unconditional Las Vegas $O(d \log d)$ on KS). *Under the KS ensemble, the multi-start Pandrosion-T_2/K=1 scheme with Strategy B' starts and the ELS fallback (Definition 4.1) admits the unconditional bound*

$$\mathbb{E}_P[\text{cost}_{total}^{B'-T_2-ELS}(P)] \;=\; O(d \log d), \tag{8}$$

conditional only on the Plancherel-decorrelation Lemma 3.2 of paper 101bis and Lemma 3.1 of this paper.

Proof. By paper 101bis, $\mathbb{E}[s^\star_{B'}] = O(1)$. By Theorem 3.2, $\mathbb{E}[N_{\text{pre-basin}}] = O(1)$. By the basin's quadratic-convergence regime, $N_{\text{basin}} = O(\log\log(1/\epsilon))$, which at $\epsilon = 10^{-12}$ contributes $O(\log d)$ epochs. The per-epoch cost is $O(d)$ (14 evaluations of P, each $O(d)$ ops). Multiplying:

$$\mathbb{E}[\text{cost}_{\text{total}}] \;=\; \underbrace{O(1)}_{s^\star_{B'}} \cdot \underbrace{(O(1) + O(\log d))}_{N_{\text{pre}}+N_{\text{basin}}} \cdot \underbrace{O(d)}_{\text{cost / epoch}} \;=\; O(d \log d).$$

□

Remark 6.2 (Comparison with Smale-17 lower bound). *Any algorithm reading P requires $\Omega(d)$ BSS operations. Theorem 6.1 achieves $O(d \log d)$, within $\log d$ of optimal. The $\log d$ gap is the basin's $O(\log\log(1/\epsilon))$ for $\epsilon = 10^{-12}$; in BSS-precision-independent statements this would be $O(d \log\log d)$, within $\log\log d$ of optimal.*

Remark 6.3 (Honest scope). *Theorem 6.1 is a Las Vegas average-case bound on the KS ensemble, not a worst-case complexity statement, and not a complexity-theoretic improvement upon Beltrán–Pardo [8] or Lairez [10], who together resolve Smale's 17th problem unconditionally. The contribution is to identify the missing super-linear pre-basin mechanism for the Pandrosion-T_2/K=1 scheme specifically.*

7 Conclusion

The empirical $\Theta(d^{0.84})$ scaling of $N_{\text{pre-basin}}$ documented in paper 101 was *not* structural to the Pandrosion-$T_2/K{=}1$ scheme. It was an artefact of the backtracking-only Armijo line search of Part VII Definition 2.1, which cannot amplify the Newton step beyond the linearisation neighbourhood and therefore requires $\Theta(\sqrt{d})$ steps to traverse the typical pre-basin gap on KS. Adding forwardtracking $\tau \in \{2^1, 2^2, \ldots, 2^6\}$ to the original $\tau \in \{2^0, 2^{-1}, \ldots\}$ closes the gap in $O(1)$ epochs with probability $\geq 1 - O(1/d)$ on KS.

The four-paper algorithmic series of Universitas Pandrosion now stabilises at:

- **Unconditional on KS:** $\mathbb{E}[\text{cost}_{\text{total}}] = O(d \log d)$ (Theorem 6.1), modulo two structural lemmas: Plancherel decorrelation (paper 101bis Lemma 3.2) and the optimal-τ scaling (Lemma 3.1).
- **Lower bound:** $\Omega(d)$ to read P.
- **Gap to optimal:** $\log d$, attributable to basin precision-tightening.

Recommendation. In all future work in this series, replace the Armijo backtracking of Part VII Definition 2.1 by the extended line search of Definition 2.1. The change is two lines of code; the speedup is $\Omega(d/\log d)$ on KS.

The reframing remains sober: this is a derivative-free root-finding heuristic with a strong empirical bound, not a new resolution of Smale's 17th problem – which was already resolved unconditionally by Lairez [10] and probabilistically by Beltrán–Pardo [8].

References

[1] I. Besevic, *Pandrosion Operator and Smale's 17th Problem.* Universitas Pandrosion, Part I (2026).

[2] I. Besevic, *Universitas Pandrosion: Part VII — A derivative-free adaptive Pandrosion scheme with non-holomorphic fallback.* Preprint, 2026.

[3] I. Besevic, *Universitas Pandrosion: Part VIII — The Geometric-Decreasing Start Strategy.* Preprint, 2026.

[4] I. Besevic, *Universitas Pandrosion: Paper 101 — Refuting and Refining the ABD Conjecture.* Preprint, 2026.

[5] I. Besevic, *Universitas Pandrosion: Paper 101bis — The KS-Adaptive Start Radius and the Phase-Transition Elimination Theorem.* Preprint, 2026.

[6] I. Besevic, *Universitas Pandrosion: Paper 101ter — The Armijo Amortised Theorem.* Preprint, 2026.

[7] I. Besevic, *Universitas Pandrosion: Paper 101quater — The Extended Line Search Rescues ABD.* Preprint, 2026.

[8] C. Beltrán and L. M. Pardo, *Smale's 17th problem: Average polynomial time to compute affine and projective solutions.* J. Amer. Math. Soc. **22** (2009), 363–385.

[9] A. Edelman and E. Kostlan, *How many zeros of a random polynomial are real?.* Bull. Amer. Math. Soc. **32** (1995), 1–37.

[10] P. Lairez, *A deterministic algorithm to compute approximate roots of polynomial systems in polynomial average time.* Found. Comput. Math. **17** (2017), 1265–1292.

[11] M. Shub and S. Smale, *Complexity of Bézout's theorem I: Geometric aspects*. J. Amer. Math. Soc. **6** (1993), 459–501.

[12] S. Smale, *Newton's method estimates from data at one point*. The Merging of Disciplines, Springer (1986), 185–196.

[13] S. Smale, *Mathematical Problems for the Next Century*. The Mathematical Intelligencer, 20(2) (1998), 7–15.